INTEGRATED PROJECT REPORT

On

PROPREP

Submitted in partial fulfillment of the requirement for
the Course IP (22CS203) of

COMPUTER SCIENCE AND ENGINEERING
B.E. Batch-2022

in

Jan -2025

Submitted By

Under the Guidance of

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CERTIFICATE

This is to be certified that the project entitled “ProPrep” has been submitted for the Bachelor of Computer Science Engineering at Chitkara University, Punjab during the academic semester January 2025 - June 2025 is a bona fide piece of project work carried out by Nishtha 2210991992, Pragti Gupta 2210992056 and Priya Gupta 2210992096 towards the partial fulfillment for the award of the course Integrated Project (CS203) under the guidance of Mr. Kamal Saluja and supervision.

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CANDIDATE'S DECLARATION

We, Nishtha 2210991992, Pragti Gupta 2210992056, Priya Gupta 2210992096, B.E.-2022 of the Chitkara University, Punjab hereby declare that the Integrated Project Report entitled **"ProPrep"** is an original work and data provided in the study is authentic to the best of our knowledge. This report has not been submitted to any other Institute for the award of any other course.

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ACKNOWLEDGEMENT

It is our pleasure to be indebted to various people, who directly or indirectly contributed to the development of this work and who influenced my thinking, behavior and acts during the course of study.

We express our sincere gratitude to all for providing me an opportunity to undergo Integrated Project as the part of the curriculum.

We are thankful to Mr. Kamal Saluja for his support, cooperation, and motivation provided to us during the training for constant inspiration, presence and blessings.

We also extend our sincere appreciation to ***Mr. Kamal Saluja*** who provided his valuable suggestions and precious time in accomplishing our integrated project report.

Lastly, we would like to thank the almighty and our parents for their moral support and friends with whom we shared our day-to day experience and received lots of suggestions that improve our quality of work.

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1. Abstract/ Keywords

Abstract: ProPrep is an AI-powered career preparation platform designed to assist job Qseekers in enhancing their interview skills. It integrates an AI Mock Interview App, which provides real-time interview simulations with intelligent feedback. Developed using React, Tailwind CSS, Clerk, Next.js, Drizzle ORM, and Gemini AI. ProPrep ensures a seamless user experience with a responsive and intuitive UI. The platform streamlines the job preparation process by offering interactive mock interviews, and insightful analytics. By leveraging AI-driven insights, ProPrep aims to empower users with the tools they need to succeed in the competitive job market.

2. Introduction to the project

In today's competitive job market, candidates often struggle with preparing for interviews effectively. ProPrep is an innovative AI-powered platform designed to bridge this gap by offering a comprehensive career preparation solution. It has an essential tool: an AI Mock Interview App, which provides real-time interview simulations with AI-driven feedback. Built using React, Tailwind CSS, Clerk, Next.js, Drizzle ORM, and Gemini AI, ProPrep ensures a seamless and user-friendly experience. By leveraging artificial intelligence, the platform provides personalized interactive interview practice, and insightful analytics to enhance job seekers' confidence and readiness. ProPrep is designed to be a one-stop solution for individuals looking to improve their employability and secure their dream jobs efficiently.

2.1 Background

With the increasing competition in the job market, job seekers often face challenges in preparing effectively for interviews. Traditional methods of interview practice lack personalization and fail to provide real-time feedback. To address these issues, ProPrep was developed as an AI-powered platform that enhances the job preparation process. The project integrates an AI Mock Interview App, which provides realistic interview simulations with AI-driven insights. Utilizing advanced technologies such as React,

Tailwind CSS, Clerk, Next.js, Drizzle ORM, and Gemini AI, ProPrep delivers a seamless and interactive user experience. The platform aims to empower job seekers with personalized career guidance, making the transition from job search to employment smoother and more efficient.

2.2 Problem Statement

Job seekers often struggle with critical aspects of career preparation: gaining effective interview experience. Interview preparation is often inconsistent and lacks real-time feedback, making it difficult for candidates to identify and improve their weaknesses. There is a need for an AI-driven solution that can provide interactive mock interview simulations to enhance job seekers' confidence and readiness. ProPrep addresses this gap by integrating an AI Mock Interview App, leveraging React, Tailwind CSS, Clerk, Next.js, Drizzle ORM, and Gemini AI to deliver a personalized, efficient, and user-friendly job preparation experience.

3. Software and Hardware Requirement Specification

3.1 Methods

Backend Development:

- User Authentication**:** Clerk authentication was integrated to ensure secure user login and session management.
- API Development**:** Next.js with API routes was used to handle user management, interview scheduling, and AI interactions.
- AI Integration**:** Gemini AI was implemented to conduct mock interviews and provide real-time feedback based on user responses.
- Data Management: Drizzle ORM was used for efficient and structured database interactions

Frontend Development:

- UI/UX Design**:** Next.js and Tailwind CSS were used to create an intuitive and responsive user interface
- Routing**:** Next.js routing was implemented for seamless navigation between different features.
- State Management: React hooks and built-in state management were utilized to maintain a smooth user experience

Database Design:

- PostgreSQL**:** A relational database was chosen for structured storage of user data, interview results, and AI feedback
- ORM Usage: Drizzle ORM was used to manage database operations efficiently.

3.2 Programming/Working Environment

- **Backend:** Node.js environment, Next.js with API routes for server-side logic, PostgreSQL for data storage, Clerk for authentication, and Drizzle ORM for database management
- **Frontend:** Next.js framework with React components, Tailwind CSS for styling, and modern UI/UX practices.
- **Tools:** VS Code for development, Postman for API testing, Git and GitHub for version control, and Prisma Studio for database inspection
- **Deployment:** The application will be deployed using Vercel for the frontend and backend, with a managed PostgreSQL database on a cloud provider.

3.3 Requirements to Run the Application

- **Node.js and npm:** Required for running the backend services and API routes.
- **PostgreSQL:** Used for structured data storage and management.
- **Next.js:** Framework for rendering the frontend and handling API requests.

- **Tailwind CSS:** For styling and responsive UI design.
- **Clerk Authentication:** For secure user authentication and session management.
- **Gemini AI:** Integrated for AI-driven mock interview simulations and feedback.
- **Drizzle ORM:** Used for efficient database interactions.

4. Database Analyzing, Design and Implementation

The PostgreSQL database schema was designed around key entities, including users, interviews, responses, AI feedback, and session logs. Each table (e.g., users, interviews, responses) holds the necessary fields required to perform efficient CRUD operations while ensuring data integrity and scalability. Drizzle ORM was used to manage database interactions effectively.

The database was structured to support:

• **User profiles** (storing candidate details, authentication data, and session history).

• **Interview sessions** (containing AI-generated questions, user responses, and timestamps).

• **AI feedback** (storing AI-driven evaluations and improvement suggestions).

• **Session logs** (tracking past mock interviews and user progress).

5. Program's Structure Analyzing and GUI Constructing

The program structure is divided into:

- Backend: Handles authentication, API routes and database queries.
- Frontend: Manages user access, analytics, and system settings. Accesses AI-powered interview sessions, receives feedback, and tracks progress.

The GUI is designed with Tailwind CSS to ensure responsiveness across all devices.

The main features include Dashboard, AI Interview Interface and Feedback System.

6. Code-Implementation and Database Connections

- Code Implementation: AI Interview Mocker project follows a modular approach, ensuring scalability and maintainability. The backend services are developed using Next.js API routes, integrating Gemini AI for generating interview responses and feedback. Clerk is implemented for secure authentication and authorization. The frontend, built with Next.js and Tailwind CSS, provides an intuitive interface for users to participate in AI-driven mock interviews. API routes facilitate user authentication, interview session handling, feedback generation, and ensuring seamless communication between the frontend and backend.
- Database Connections: The database connections are established using Drizzle ORM with PostgreSQL, ensuring efficient and structured data management. The database stores user profiles, interview session data, AI-generated feedback, and payment transactions. API endpoints interact with PostgreSQL using Drizzle ORM queries, enabling secure and optimized data retrieval and manipulation based on user interactions. The system ensures data consistency, high availability, and security while managing user sessions and interview history effectively.

7. Limitations

- **Live Video Interview Feature:** Currently, the platform only supports text-based AI mock interviews. The integration of real-time video interviews using WebRTC or third-party services like ZegoUIKit or Twilio is under development to enhance the realism of mock interview sessions.
- **Limited Customization of AI Responses:** While Gemini AI provides structured feedback, users have limited control over customizing interview difficulty levels, industry-specific questions, or tailored feedback formats. Future updates will introduce more personalization features to align interviews with specific job roles and experience levels.
- **No Multi-Language Support:** At present, the system supports only English for interview sessions and feedback generation. Plans are in place to integrate multi-language support to cater to a more diverse user base.

8. Conclusion

The development of the AI Interview Mocker project has made significant progress, with core functionalities operating as intended. The backend is stable and efficiently manages user authentication, AI-driven mock interviews, and feedback generation. The frontend is fully responsive and provides an intuitive user interface, ensuring a seamless experience for candidates engaging in AI-powered interview sessions.

However, certain features are still under active development. The integration of a live video interview feature is in progress, which will enhance the realism of mock interview experiences by allowing real-time interactions with AI or human interviewers. Additionally, improvements in customizing AI responses and supporting multi-language interviews are planned to make the platform more inclusive and personalized. Expanding payment gateway options beyond Stripe is also on the roadmap to accommodate a wider user base.

Despite these ongoing developments, the platform has successfully laid a strong foundation with its current capabilities. Once the pending features are fully implemented, the AI Interview Mocker will provide a comprehensive, interactive, and intelligent mock interview experience, helping users prepare effectively for job interviews with real-time AI assistance.

9. Future Scope

- **Personalized AI Feedback & Reports:** Enhancing AI feedback with detailed performance analytics, improvement suggestions, and custom learning paths will provide users with more actionable insights.
- **Resume Analysis & Job Recommendations:** Integrating an AI-powered resume analyzer and job recommendation engine will help users optimize their resumes and find job opportunities aligned with their skills and experience.
- **Personalized AI Feedback & Reports:** Enhancing AI feedback with detailed performance analytics, improvement suggestions, and custom learning paths will provide users with more actionable insights.

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